



B.Tech. Degree IV Semester Examination April 2017

ME 15-1404 APPLIED THERMODYNAMICS

(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain briefly the limitations of first law of thermodynamics.
- (b) State and prove Carnot's theorem.
- (c) Derive equation for entropy change for reversible polytropic process and isothermal process.
- (d) Explain basic components of a steam power plant.
- (e) What are the desirable characteristics of a working fluid in a vapour power cycle?
- (f) Describe super saturated flow in a steam nozzle.
- (g) Explain the effect of variation of back pressure in a steam nozzle.
- (h) Write a short note on effects of friction on blades of a steam turbine.
- (i) Differentiate between volumetric and gravimetric analysis of gas mixtures.
- (j) Explain gas constant of gas mixtures.

PART B

(4 × 10 = 40)

- II. (a) Derive steady flow energy equation and apply the SFEE to flow through a nozzle, diffuser, turbine, compressor and throttling device. (5)
 - (b) A centrifugal air compressor delivers 900 kg/hr of air. Air velocity at inlet and outlet are 5 m/s and 7.5 m/s. Specific volume at inlet and outlet are 0.5 m³/kg and 0.15 m³/kg. Enthalpy of inlet air and compressed air are 5 kJ/kg and 20 kJ/kg. Heat lost to inter-stage cooling is 75.6 kJ/s. Compute the motor power required to drive the compressor and the ratio of inlet to outlet pipe diameter. (5)
- OR**
- III. (a) Derive Maxwell equations and T.ds equations from fundamentals. (6)
 - (b) A cold storage unit of 30 ton of refrigeration capacity operates between 263 K and 303 K. Determine the monthly cost of running the plant if it is run for 16 hours a day and the plant has half of the COP of a Carnot cycle. [Take 30 days a month, cost of power = ₹5.25 per kW-hr, 1 ton of refrigeration = 3.5 kW]. (4)
- IV. (a) Describe Reheat cycle and compare it with simple Rankine cycle. (5)
 - (b) A steam power plant operates in a Rankine cycle with superheated steam. The inlet steam conditions are 100 bar, 550°C. The condenser operating pressure is 0.05 bar. Determine the efficiency of the cycle. (5)

OR

(P.T.O.)

- V. (a) Describe briefly the mercury-water binary cycle. (4)
 (b) A simple Rankine cycle works between pressure 28 bar and 0.06 bar, the initial condition of steam being dry saturated. Calculate the cycle efficiency, work ratio and specific steam consumption. (6)

- VI. (a) Derive an expression for the throat pressure for maximum discharge in a steam nozzle. (5)
 (b) Dry saturated steam at 10 bar and an initial velocity of 90 m/s enters a steam nozzle and is discharged at 1.5 bar. Determine (i) final velocity of steam (ii) if the friction loss is 15% of the total enthalpy drop, find the final velocity and percentage reduction in velocity of steam. (5)

OR

- VII. (a) Write a short note on pressure compounding of an impulse turbine. (4)
 (b) The velocity of steam at inlet to a simple impulse turbine is 1000 m/s and the nozzle angle is 20° . The mean blade speed is 400 m/s and the blades are symmetrical. The mass flow rate of steam is 0.75 kg/s. The friction effects on the blade are negligible. Estimate (i) the blade angle (ii) the tangential force on the blades (iii) the axial thrust (iv) diagram power. (6)

- VIII. (a) Derive an expression for the final temperature in adiabatic mixing of streams. (4)
 (b) Two vessels, both containing nitrogen are connected by a valve which is opened to allow the contents to mix and achieve an equilibrium temperature of 30°C . Before mixing the composition is as follows: (6)

Vessel	Pressure	Temperature	Contents
A	15 bar	60°C	0.5 kg
B	6.5 bar	25°C	3.5 kg

If the vessel is perfectly insulated find: (i) the final temperature (ii) the final pressure.

OR

- IX. (a) Derive an expression for the apparent molecular mass of a mixture of gases. (4)
 (b) The volumetric composition of a sample of gas is as follows: 10% hydrogen, 20% methane, 5% carbon dioxide 15% carbon monoxide and 50% nitrogen. Determine the gravimetric composition, apparent molecular mass of the gas and the partial pressure of each constituent if the total pressure is 1.013 bars. (6)